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# Investigation on the effect of vertical baffle plate on droplet size distribution and mixing efficiency in Taylor Reactor

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ABSTRACT

The Taylor reactor is a dynamic mixer based on the principles of Taylor vortex flow. However, traditional Taylor reactors feature smooth inner and outer cylinder surfaces, presenting challenges in achieving uniform mixing and efficient mass transfer. These limitations do not meet the requirements of modern chemical processes for highly efficient mixing equipment. This article introduces a novel Taylor reactor equipped with vertical baffle plates, designed to overcome these challenges. A comprehensive methodology combining CFD-PBM and experimental techniques was employed to investigate both droplet size distribution and mixing performance in the annular gap. Compared to traditional designs, the new device demonstrates significantly lower *COV* at 0.018 and Sauter mean diameter at 38.3 μm. Optimal mixing efficiency is achieved at rotating Reynolds number ($Re$) of $1.17 \times 10^5$ and inlet Reynolds number ($Re_i$) of $2.65 \times 10^3$. Numerical simulation results suggest that vertical baffles effectively enhance liquid-liquid interactions including collision, interface contact, and droplet residence time within the annular gap. Moreover, increased turbulent stress in the flow field contributes to greater vorticity, turbulence intensity, and turbulent kinetic energy resulting in smaller droplet sizes and improved mixing efficiency. These findings provide a theoretical foundation for hydraulic structural optimization design as well as widespread application of Taylor reactors.

## 1. Introduction

Extraction, a process combining mixing and separation, is directly influenced by the quality of mass transfer. To achieve efficient liquid-liquid mixing performance, it is necessary for the dispersed phase should have a narrow droplet distribution of size and be evenly dispersed throughout the continuous phase [1,2]. In recent years, researchers have extensively researched the mechanisms and performance of mixing [3-11]. As a dynamic mixer, the Taylor reactor can generate Taylor vortices when its inner cylinder rotates at high speed. As a result, it has been widely applied in fields such as chemical engineering [12,13], materials science, biology [14], medicine [15-18], and environmental applications. The rapid development of modern industry has led to the emergence of higher performance requirements for intensifying mixing processes, thereby demanding exceptional mixing effects. This presents a substantial challenge for traditional Taylor reactors. Particularly under conditions where the phase that is continuous and scattered exhibit a high ratio and high viscosity, the scattered phase's droplet size can reach millimeter or even micrometer levels. Additionally, high continuous phase velocities further degrade the mixing efficiency of conventional liquid-liquid mixing device, increasing both equipment size and operational costs in industrial applications [19]. Therefore, optimizing Taylor reactor structure and developing new types of Taylor reactors hold significant research value and significance for enhancing mixing efficiency.

Structural optimization of Taylor reactors typically focuses on two key aspects: the size and shape of the annular gap. For Taylor reactors, the diameter ratio (inner diameter to outer diameter) and aspect ratio (reactor height to annular gap width) are crucial parameters that define the flow characteristics within the annular gap. Therefore, most research on Taylor reactors ensures that their structural dimensions meet these two conditions [20-22]. In recent years, some researchers have investigated incorporating internal components or modifying the shape of the annular gap to improve mixing performance. Szczechowski et al. [23]

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demonstrated that the introduction of horizontal baffles does not alter the axisymmetric nature of Taylor-Couette flow; instead, they provide additional surfaces to restrict vortex flows while horizontal baffle plates affect vortex sizes. Deshmukh et al. [24] investigated different flow patterns with or without bottom blades in the annular gap, finding that bottom blades reduce vortex formation. Deng et al. [25] examined Taylor vortical flow between a rotating inner cylinder and a fixed outer cylinder, finding that vertical baffles disrupt axisymmetric vortices, leading to three-dimensional flow patterns. To address the issues of low velocity at the core of Taylor vortices and weak shear force in traditional Taylor reactors, Wiener [26] designed a leaf-shaped cross-section for a Taylor reactor which exhibited narrower shear force distribution due to periodic variation in its annular gap width through CFD numerical simulation analysis; thus, eliminating disadvantages associated with low fluid shear rate in conventional designs while enhancing interphase mass transfer processes. Although numerous studies focus on enhancing the structure of Taylor reactors; however, research is still primarily focused on aspects such as aspect ratio, longitudinal-to-transverse ratio, fluid flow dynamics, and wall heat transfer optimization. Insufficient attention has been paid to the impact of wall structure and droplet dispersion characteristics or exploring how two-phase flow patterns within turbulent Taylor vortices affect droplet size distribution. Although some of the aforementioned studies have demonstrated improvements in mixing performance for Taylor reactors, existing literature has yet to identify an effective method to address uneven mixing within the annular gap under high phase ratios [27,28].

To effectively enhance mixing performance, it is essential to study flow properties numerically, including form of flow zones and Taylor vortex vorticity, especially in the presence of baffle structures [29,30]. While some visualization experiments [31] provide methods for obtaining macroscopic parameters of liquid-liquid mixing flow, limitations in measurement technology make it difficult to acquire quantitative data on two-phase interface interaction, droplet motion behavior and dispersed phase flow distribution. Conversely, computational fluid dynamics (CFD) provides a practical and precise way to quantitatively describing flow field structures. The main factors influencing liquid-liquid mixing efficiency include dispersed phase concentration, droplet size distribution (DSD), constant phase speed zone and development of vorticity [32]. Many studies have employed CFD used together population balance model (PBM) simulation methods to accurately explain the distributed phase's DSD [33-35]. Chen et al. [36] employed CFD-PBM numerical simulation to establish a pilot-scale stirred bioreactor, examining the effects of impeller configuration, aperture size, and gas distributor structure on reactor performance. This method applies not only to liquid-liquid mixing systems but also to simulating polymer kinetics in tailings slurry thickening process within feed wells, effectively addressing the issue of solid particle flocculation affecting concentration [37]. Qin et al. [38] utilized an EMMS theory-corrected CFD-PBM model to predict the evolution of DSD in weakly agglomerated emulsion systems within Megatron rotor-stator mixing device and achieved excellent agreement between experimental data and simulation results. The CFD-PBM model has proven to be mature and reliable for simulating the dispersed phase DSD for achieving uniform mixing. However, there is currently a lack of research reports regarding the application of CFD-PBM models in optimizing the structure of Taylor reactors, and the underlying mechanism responsible for enhancing mixing remains unknown.

Therefore, this article proposes a design methodology aimed at reducing the droplet diameter of the dispersed phase and enhancing the uniformity of its distribution by incorporating vertical baffle plates in the annular gap of Taylor reactors. The aim is to improve homogeneity and mixing efficiency during liquid-liquid mixing within the annular gap. Additionally, this study systematically examines the correlation between rotating Reynolds number (*Re*) and droplet size distribution as well as coefficient of variation (*COV*). Using a CFD-PBM coupled model, we simulate and predict macroscopic flow field characteristics of an improved Taylor reactor. We establish a relationship within droplet fragmentation characteristics and turbulent shear stress, revealing how baffle structure enhances mixing performance. These findings give a conceptual framework for hydraulic structural optimization procedure as well as widespread application of Taylor reactors.

## 2. Materials and method

### 2.1. Geometric model

To enhance the mixing performance of Taylor reactors, this study initially designed two types of vertical Taylor reactors: one with a smooth annular gap and the other incorporating vertical baffles, adhering to the design criteria for Taylor reactors. Considering that the experiment involved mixing a light oil phase with a heavy water phase, the study focused on optimizing the structure of a bottom-entry Taylor reactor. Key design parameters include the barrate nominal diameter ($D_i$), outer cylinder diameter ($D_o$), working length ($L$), annular gap width ($d$), diameter ratio ($\eta$) and aspect ratio ($\Gamma$). The operational principle of the Taylor reactor primarily relies on high-speed rotation of the inner cylinder to generate Taylor vortices in the annular gap. The generation of vortex flow depends on two parameters: $\eta$ and $\Gamma$. The definitions of $\eta$ and $\Gamma$ are as follows:

$$\eta = D_i / D_o \tag{1}$$

$$\Gamma = L/d \tag{2}$$

Snyder [22] suggested that neglecting the end effect is beneficial for maintaining stable Taylor vortices when $\Gamma > 10$. To avoid oscillatory flow, $\eta$ is typically designed to be greater than 0.71 when it falls below this threshold. In this study, the utilized Taylor reactors were designed with $\eta$ of 0.83 and $\Gamma$ of 15. Fig. 1(a) depicts both the no baffle Taylor reactor (NTR) and vertical baffle Taylor reactor (VBTR), emphasizing specifically on the installation of vertical baffles on the inner wall of the shell in this investigation.

For the experiment, the Taylor reactor's annular gap was designed in two configurations: without baffle plate and with vertical baffle plates. The shell was constructed using organic glass material, while the barrate is made of stainless steel. In the NTR, the inlets for light and heavy phases were symmetrically positioned along the shell's tangent line at the lower part, while the mixed-phase outlet was located along the tangent line at the upper part. The VBTR is distinguished by incorporating vertical baffle plates on its inner wall, sharing identical overall dimensions as NTR. Table 1 presents key parameters for structural design of Taylor reactor, whereas Fig. 1(b) illustrates dimensional design of VBTR structure.

### 2.2. Experimental process

#### 2.2.1. Reagents for experimental purposes

This study utilizes a kerosene-water mixture for experimentation, with water serving as the constant stage and kerosene acts as the dispersed phase (with a concentration ranging from 1 to 10 %). The physical properties of each substance utilized in the experiment are presented in Table 2.

#### 2.2.2. Experimental device

Fig. 2 illustrates the process used to characterize the dispersibility performance of the liquid-liquid mixing device. The dispersed phase fluid is delivered to the inlet-1 of the Taylor reactor using a peristaltic pump, while the continuous phase fluid is delivered to the inlet-2 through another peristaltic pump. The feed flow rate can be precisely adjusted via the control panel of the peristaltic pump. A brushless motor with a speed range of 100 to 3000 rpm that meets the operational speed requirements for Taylor reactor applications. After mixing within the annular gap region of the Taylor reactor, both phases exit from an upper

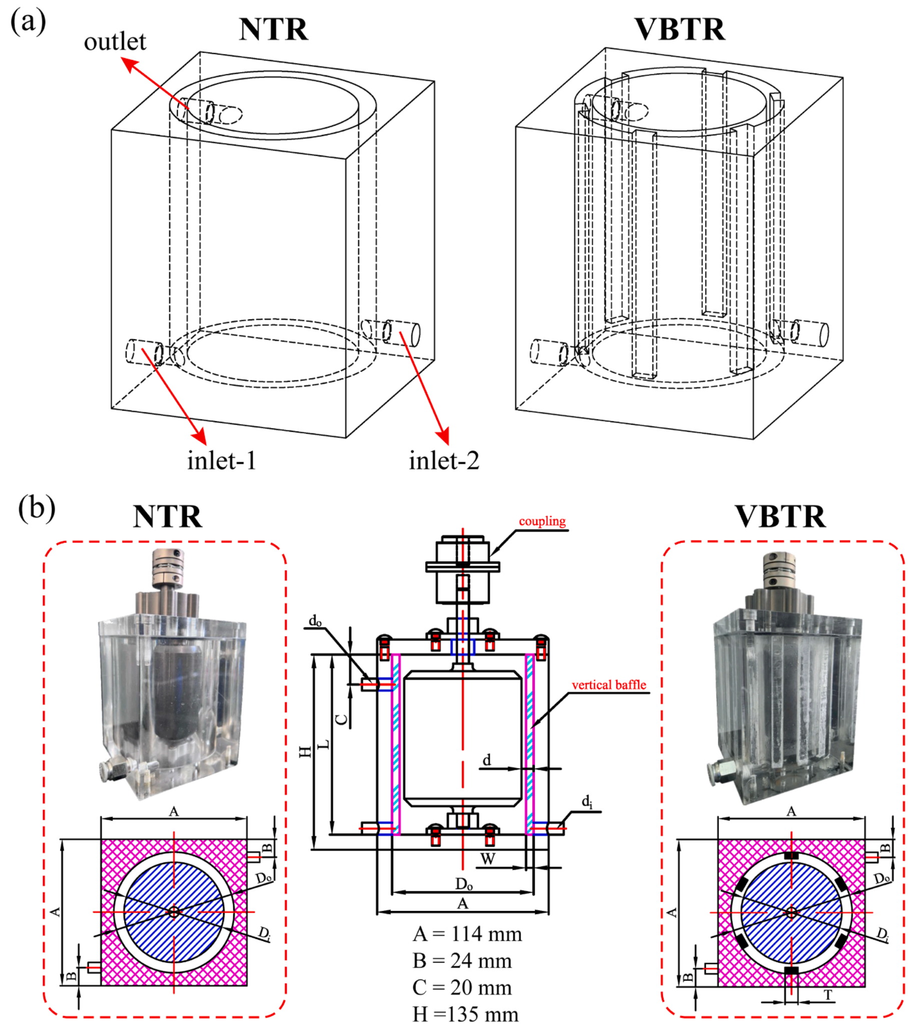


**Fig. 1.** (a) The NTR and VBTR models and (b) Schematic diagram of Taylor reactor.

**Table 1**
Designed parameters of Taylor reactor geometry.

| Parameter | Symbol | Dimension | Unit |
|---|---|---|---|
| Barrate nominal diameter | $D_i$ | 78 | mm |
| Outer cylinder diameter | $D_o$ | 94 | mm |
| Working length | $L$ | 120 | mm |
| Annular gap width | $d$ | 8 | mm |
| Vertical baffle width | $W$ | 5 | mm |
| Vertical baffle thickness | $T$ | 8 | mm |
| Inlet nozzle diameter | $d_i$ | 8 | mm |
| Outlet nozzle diameter | $d_o$ | 8 | mm |
| Vertical baffle quantity | $n$ | 6 | / |

**Table 2**
Experimental fluid physical properties parameters (room temperature 20 °C).

| Fluid | Specification | Density/kg·$m^{-3}$ | Viscosity/mPa·s |
|---|---|---|---|
| Kerosene | AR | 801.2 | 2.500 |
| Ultra-pure water | AR | 998.2 | 1.003 |

outlet and are directed into a storage tank. For studying liquid-liquid dispersion performance, an analyzer of laser size of particles (BT-9300ST, China Dandong Baxter Instrument Co., Ltd.) was utilized to measure droplet size distribution in samples collected from outlet fluids. Additionally, NTR were used as benchmarks. To guarantee the accuracy of the data throughout all experiments, each experimental condition was measured three times, and the results were averaged. Additionally, to mitigate potential data inaccuracies arising from coalescence, a uniform sample volume of 60 ml was consistently employed in the laser particle size analyzer for all trials. The residence time during analysis was maintained at approximately 30 s to ensure experimental consistency.

To evaluate the mixing efficiency of two types of Taylor reactors, this study focused on examining the variation in droplet size under different rotating speeds of barrate ($N$), feed rates, and dispersed phase concentrations. The feed rate is quantified using the inlet Reynolds number ($Re_i$), defined as:

$$Re_i = \rho_c u d_i / \mu_c \tag{3}$$

where $\mu_c$ represents the viscosity of the continuous phase, $\rho_c$ represents the density of the continuous phase, and $u$ signifies the mean flow rate in the inlet. Based on the applicable range of mixing flux for a φ80 Taylor reactor, this study determined that $Re_i$ should fall within $1.06 \times 10^3$ to $4.24 \times 10^3$ for experimental conditions.

### 2.3. *Evaluation methodology*

#### 2.3.1. *Mixing uniformity*

The coefficient of variation (*COV*) is employed in this study to better capture the uniformity of device mixing, serving as an indicator for evaluating the mixing performance of Taylor reactors [39,40]. It quantifies the level of mixing between two distinct fluids. Specifically, *COV* is

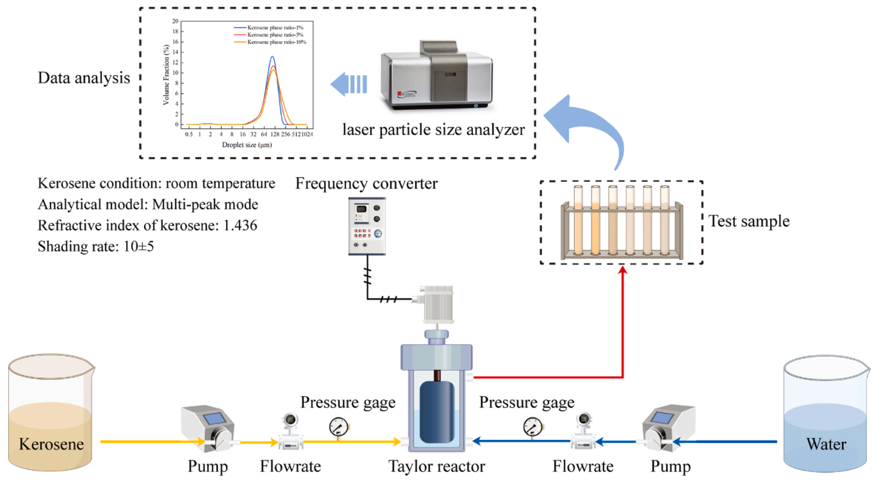


**Fig. 2.** Experimental process of the droplet size distributions in Taylor reactor.

calculated by dividing the standard deviation ($\sigma$) by the average concentration ($\overline{C}$) within a specific cross-section in the mixing region.

$$COV = \frac{\sigma}{\overline{C}} \tag{4}$$

The standard deviation ($\sigma$) of the sample concentration is determined by employing the subsequent Eq. (5):

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(C_i - \overline{C})^2}{N-1}} \tag{5}$$

The average concentration ($\overline{C}$) is calculated using the following Eq. (6):

$$\overline{C} = \frac{1}{N}\sum_{i=1}^{N} C_i \tag{6}$$

The equation defines $C_i$ as the dispersed phase's level in a specific location on a cross-section of a mixing section, representing the number of samples collected at that location. *COV* is a macroscopic parameter ranging from 0 to 1. A lower *COV* value indicates smaller concentration variations between sampling points and higher uniformity in mixing. It is generally accepted that when $COV \leq 0.1$, complete mixing can be achieved; whereas, when $COV > 0.1$, it signifies complete non-uniform mixing [40].

### 2.3.2. Droplet size distribution

The experiment's droplet size is characterized by a discrete distribution of droplet volume density, which can be mathematically represented as Eq. (7) [41]:

$$q_{3,i} = \frac{f_i}{d_i} \tag{7}$$

In the equation, $f_i$ represents the volumetric fraction occupied by each individual droplet size, calculated using the following Eq. (8):

$$f_i = \frac{N_i \frac{\pi}{6} d_i^3}{\sum_{j=1}^{M} N_j \frac{\pi}{6} d_j^3} \tag{8}$$

The sizes of dispersed phase droplets are categorized as $i, j = 1, \cdots\cdots, M$, where $d_i$ represents the number $i$ droplet's diameter. There are primarily two methods for measuring the size of dispersed phase droplets: image measurement method [42] and remote measurement method [43]. To simplify subsequent image while reducing the data analysis workload, this study selects four characteristic values to describe the size distribution, namely $d_{10}$, $d_{50}$, $d_{90}$ and $d_{3,2}$. The average particle size commonly used for Sauter mean diameter ($d_{3,2}$) is considered as a representative evaluation index. Its expression typically using the following Eq. (9):

$$d_{3,2} = \left(\sum_{j=1}^{M}\frac{f_j}{d_j}\right)^{-1} = \frac{\sum_{i=1}^{n} n_i d_i^3}{\sum_{i=1}^{n} n_i d_i^2} \tag{9}$$

The equation defines $n_i$ as the number of droplets with particle sizes ranging from $d_i$ to $d_{i+1}$. Among them, $d_{90}$ represents the particle size corresponding to 90 % cumulative particle count, while $d_{10}$ and $d_{50}$ are defined similarly.

## 3. Numerical simulation

### 3.1. CFD-PBM model

The fluid flow characteristics within the annular gap of a Taylor reactor vary across different flow regimes, with the anisotropic motion of multi-stage Taylor vortices manifests as a rapidly evolving dynamic wave-like turbulence. The Reynolds Stress Model (RSM), which is founded on Reynolds averaging. Previous studies have demonstrated the applicability of RSM in analyzing Taylor vortices and eddies [20,44]. This study employs the RSM model for turbulence simulation in the CFD analysis.

The existing techniques for calculating multiple-phase flow mainly consist of the Eulerian and Lagrangian approaches. The Eulerian method is widely used to simulate blending and dividing phenomena in multiple-phase flows, treating every stage as a continuous medium that can interpenetrate [45]. The Euler-Euler model, which solves the momentum equation and continuity equation separately for any phase via combining pressure and velocity models. The Euler-Euler model has been selected as the multiphase flow model in this study. The Schiller & Nauman drag model has been selected to accurately represent hydrodynamic forces in the Euler-Euler model.

The CFD-PBM model accurately captures the spatiotemporal evolution of droplet size distribution within a flow field [46-48]. For a specific number density distribution $f(\xi)$ of droplets with volume $\xi$, the transport equation that the particle group equilibrium model, as shown in Eq. (10), can be derived. Given that mass transfer is assumed to be negligible, no growth term is incorporated in Eq. (10):

$$\frac{\partial f(\xi)}{\partial t} + v\nabla f(\xi) = B_c(\xi) - D_c(\xi) + B_b(\xi) - D_b(\xi) \tag{10}$$

In the droplet coalescence process, the birth rate of daughter particles and the death rate of mother particles are defined by Eq. (11) and

Eq. (12) as follows:

$$B_c(\xi) = \frac{1}{2}\int_{\xi_{\min}}^{\xi} \omega(\xi', \xi-\xi')f(\xi')f(\xi-\xi')d\xi' \tag{11}$$

$$D_c(\xi) = f(\xi)\cdot\int_{\xi_{\min}}^{\xi_{\max}} \omega(\xi', \xi)f(\xi')d\xi' \tag{12}$$

Likewise, in the droplet breakup process, the birth rate of daughter particles and the death rate of mother particles are defined by Eq. (13) and Eq. (14) as follows:

$$B_b(\xi) = \int_{\xi}^{\xi_{\max}} m(\xi)\beta(\xi', \xi)g(\xi')f(\xi')d\xi' \tag{13}$$

$$D_b(\xi) = g(\xi)f(\xi) \tag{14}$$

This study considers only binary splitting due to the absence of assumptions related to multiple splitting. Experimental results demonstrate that mother droplets are most likely to split into equally sized daughter droplets in liquid-liquid dispersion systems [49]. Consequently, this study adopts the model expression given by Eq. (15) for calculating the distribution of daughter droplets during mixing processes within an annular gap.

The distribution of daughter particles is defined as follows:

$$\beta(\xi', \xi) = 30\left(\frac{\xi}{\xi'}\right)^2\left(1-\frac{\xi}{\xi'}\right)^2 \tag{15}$$

The widely adopted Coulaloglou and Tavlaride [50] droplet coalescence and breakup model in liquid-liquid dispersion systems (expressed as Eq. (16) and Eq. (17)) has been extensively validated in existing literature [51-53]. Both kernel models assume that breaking droplets have sizes corresponding to those of coalescing droplets at a turbulent scale, which primarily depends on turbulence dissipation rate $\varepsilon$ [50]. The physical property parameters for both models, including dispersed phase volume fraction $\alpha_d$, dissipation rate of turbulence $\varepsilon$, surface tension $\sigma$, viscosity $\mu$, density $\rho$.

$$g(\xi') = C_1\xi'^{-2/3}\frac{\varepsilon^{1/3}}{1+\alpha_d}\exp\left[-\frac{C_2\sigma(1+\alpha_d)}{\rho_d\varepsilon^{2/3}\xi'^{5/3}}\right] \tag{16}$$

$$\omega(\xi, \xi') = \left[C_3\frac{\varepsilon^{1/3}}{1+\alpha_d}(\xi+\xi')^2\left(\xi^{2/3}+\xi'^{2/3}\right)^{1/2}\right]\exp\left[-\frac{C_4\mu\rho_d\varepsilon}{\sigma^2(1+\alpha_d)^3}\left(\frac{\xi\xi'}{\xi+\xi'}\right)^4\right] \tag{17}$$

Furthermore, the outcomes of simulations are affected by dimensionless constants $C_1$—$C_4$ values. To ensure consistency in mixing characteristics and form with previous studies constant reference principles suggested by Rave et al. [41], with $C_1$=0.1, $C_2$=3.2exp15, $C_3$=1.6, $C_4$=0.32 chosen as values are utilized within this investigation. When the computation first started, the distribution of particle sizes is split up into 25 distinct bins (Bin 0 ~ Bin 24), ranging from 0.02 to 0.243 mm according to the standards of minimum size and geometric proportionality. The homogeneous discretization approach is employed to set the PBM model in this study. The breakup equation in discrete method is founded on Hagesather formula.

### 3.2. Model mesh

The fluid domain model of the Taylor reactor was developed using Solidworks software, with the computational fluid domain was discretized into hexahedral meshes using ICEM CFD software, as shown in Fig. 3(a). Accurate numerical simulations are critically dependent on the accurate determination of boundary layer thickness, owing to the pronounced variations in fluid velocity near the wall. The determination of boundary layer thickness and the number of refinement layers is achieved through the calculation of $y^+$. The calculation of $y^+$ is described by the following equation:

$$y^+ = \frac{\sqrt{\tau_\omega/\rho}\cdot\Delta n}{\upsilon} = \frac{\rho u_\tau\Delta y}{\mu} \tag{18}$$

In the equation, $u_\tau$ represents the friction velocity, $\rho$ represents fluid density, $\mu$ represents fluid's dynamic viscosity, and $\Delta y$ represents the distance between the first layer grid node and the wall. Eq. (18) reveals that $y^+$ is a dimensionless ratio of friction velocity, wall distance, and kinematic viscosity. For turbulent flows with high Reynolds numbers, $y^+$ typically lies within the range of 30 to 300. Previous research has shown that when using locally averaged turbulence models, $y^+$ can be restricted to a narrower range of 20 to 80. Based on calculations, the distance between the first layer grid node and the wall was determined to be 0.9 mm. This approach allows for iterative local grid refinement in regions adjacent to barrate walls and gap edge walls.

Validating grid independence is essential for improving computational efficiency and reducing simulation errors. The vertical pressure distribution at the center of the air gap was compared across different mesh resolutions (443,212, 702,138, and 1,013,213) under identical boundary conditions, as depicted in Fig. 3(b), The pressure distributions exhibit consistency across various grid resolutions. The model employing a grid resolution of 443,212 demonstrates results differing by <2 % from those obtained using other grid sizes, indicating that the simulation outcomes are independent of grid size. To optimize computational resources and accelerate the calculations, a mesh resolution of 443,212 was ultimately selected for this study.

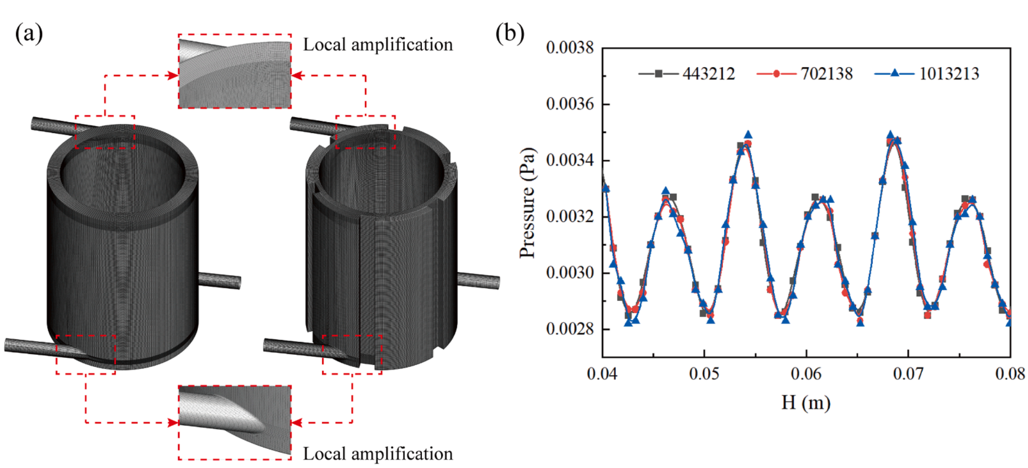


**Fig. 3.** (a) Hexahedral mesh and (b) Mesh independence verification.

### 3.3. Numerical methods and boundary conditions

The models within this investigation were simulated using the commercial ANSYS Fluent software, employing a pressure-based solver that is chosen for steady-state simulations. The SIMPLE algorithm was used to produce stress-velocity coupling, and the PRESTO discretization technique was used to calculate pressure. The discretization of volume fraction, momentum, dissipation rate, and Reynolds stress was carried out using the Quadratic Upwind Interpolation for Convection Kinematics (QUICK) scheme. The First-order Upwind scheme was applied to the kerosene bin. Custom User-Defined Functions (UDFs) were separately developed to integrate fracture and aggregation models into the Euler model algorithm. To ensure simulation accuracy, 15,000 iterations were conducted to stabilize the numerical calculation before introducing the dispersed phase.

The simulation's inlet boundary condition was established to velocity-inlet, and free outflow with a 5 % turbulence intensity is the outlet boundary condition that is established. The non-dimensional axial Reynolds number $Re_a$ is utilized to characterize the ratio of inertial force to viscous force, and its expression can be determined by the following Eq. (19):

$$Re_a = wd/\nu \tag{19}$$

Where $w$ represents the mean axial velocity across the annular gap region.

The inner wall is configured as a Moving Wall with the Y-axis serving as the rotation axis, and the rotational speed varies according to different operating conditions determined by the rotating Reynolds number ($Re$). Due to the analogy between flow in the annular gap and Taylor-Couette flow between coaxial cylinders, Taylor-Couette flow can be characterized by two dimensionless numbers, $Ta$ and $Re$, defined respectively in Eq. (20) and Eq. (21). In this paper, we select $Re$ as the representative parameter.

$$Ta = \frac{D_i(d)^3\Omega^2}{\nu^2} \tag{20}$$

$$Re = \frac{(D_i\Omega)d}{\nu} \tag{21}$$

Where $D_{\mathrm{i}}$ represents the nominal diameter of barrate, $d$ is the annular gap width, $\Omega$ is the rotating speed, and $\nu$ is the fluid kinematic viscosity.

The continuous phase is classified as water, while the scattered phase is marked as kerosene, with an interfacial tension of 0.034 N/m. Kerosene is introduced through inlet 1 while water enters through inlet 2, with a second-phase volume fraction of 1 for inlet 1 and 0 for inlet 2. The initial volume fraction of particles at Bin 0 (i.e., the maximum oil droplet size in this study is 0.243 mm) at inlet 1 is set to 1. Six radial sections (Z1-Z5 and Outlet) and one axial R-X section ($z = 0$) were selected as characteristic planes for numerical simulation analysis. To examine the relationship among the properties of the flow field and droplet size distribution in the Taylor reactor gap, three specific positions were selected for analysis. Point A is located at the lower end of gap meridian plane with coordinates (40.5, 20, 0), and additional points are sampled every 40 mm along device axis direction (refer to Fig. 4(a)). Additionally, the $d_{3,2}$ values in the numerical simulation are extracted using the "Diameter" option in the "Result" module, while the $d_{10}$ and $d_{50}$ values are obtained by exporting the average data from Bin 0 to Bin 24 and calculating them using cumulative volume fractions.

## 4. Results and discussion

### 4.1. Comparison of simulation and experiment

A comparison of the dispersed droplet sizes (represented by characteristic particle diameters $d_{10}$, $d_{3,2}$ and $d_{50}$) obtained from numerical simulations and experiments at three different rotational speeds, as depicted in Fig. 4(b), confirms the applicability of the Coulaloglou and Tavlaride droplet coalescence and breakup model for both NTR and VBTR. Since the deviation between simulation results and experimental values is within 5 %, the CFD-PBM model employed in numerical simulations is deemed reliable for both methodology and result accuracy. Additionally, it was observed that when $N < 1800$ rpm, simulated droplet size distribution is broader while above this speed threshold it becomes narrower.

### 4.2. Droplet size distributions in the experiment

#### 4.2.1. Effect of rotating speed

The variations in droplet size at different rotational speeds were examined under experimental conditions with $Re_{\mathrm{i}} = 2.65 \times 10^3$ and the dispersed phase concentration of 5 %. Fig. 5(a) presents the dispersion curves for $d_{50}$ and $d_{3,2}$ in NTR and VBTR. Both $d_{50}$ and $d_{3,2}$ decrease with increasing rotational speed. It is evident that both $d_{50}$ and $d_{3,2}$ decrease as the speed increases. Notably, VBTR consistently generates smaller droplets compared to NTR. At 1600 rpm, the $d_{50}$ of VBTR is approximately 1.8 times smaller than that of NTR, measuring 65.1 μm. This indicates that the vertical baffle Taylor reactor offers superior circumferential shear and mixing capabilities within the annular gap. Additionally, both $d_{3,2}$ curves exhibit a decreasing trend with increasing speed, consistent with Wyatt's research [54]. In VBTR, as the speed reaches 1600 rpm, there is a gradual reduction rate in droplet size

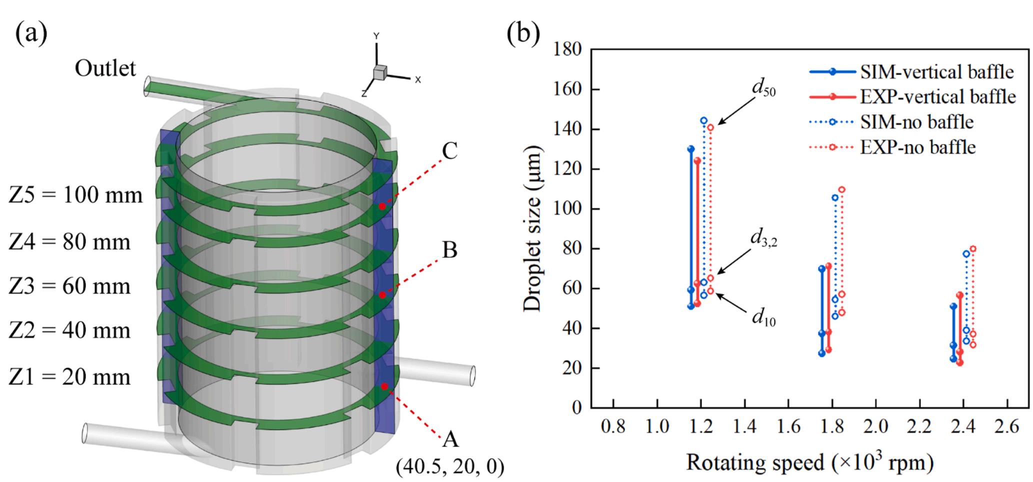


**Fig. 4.** (a) Data analysis feature planes and (b) Comparison of results between simulation and experiment.

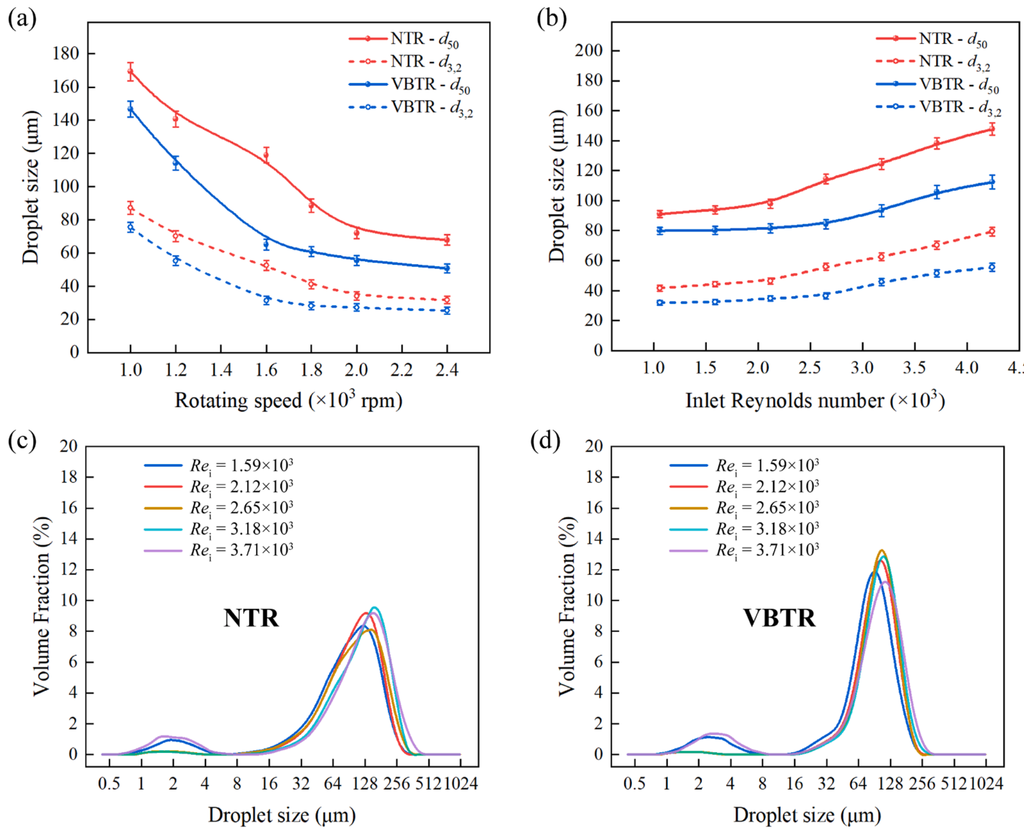


**Fig. 5.** (a) The $d_{3,2}$ and $d_{50}$ at different Rotating speed; (b) The $d_{3,2}$ and $d_{50}$ at different $Re_i$; (c) and (d) The DSD under different $Re_i$ for both Taylor reactors (data from experiments).

accompanied by a flattening curve; however, it is not until an increase to 2000 rpm that NTR shows signs of slowing down its rate of droplet size reduction. These research findings indicate that VBTR exhibits excellent shear droplet capability and is more suitable for generating smaller droplets under relatively low-speed conditions, which implies lower energy consumption and enhanced safety for industrial applications. Based on this analysis, it can be concluded that the smaller droplet sizes in VBTR result from enhanced shear forces facilitated by the vertical baffle structures within the annular gap, resulting in improved mixing performance in Taylor reactors.

### 4.2.2. *Effect of inlet's Reynolds number*

Experiments were conducted at varying $Re_i$ values for the NTR and VBTR. Fig. 5(b) displays the $d_{50}$ and $d_{3,2}$ curves of the dispersed phase under constant operating conditions with dispersed phase concentration of 5 % and rotational speed of 1800 rpm. As $Re_i$ increases, both $d_{50}$ and $d_{3,2}$ increased; however, VBTR exhibited smaller droplet sizes than NTR across all $Re_i$ conditions. For VBTR, only when $Re_i$ reached $2.65 \times 10^3$ did the droplet size exhibit rapid growth trends while for NTR, the droplet size curve almost linearly increased until it reached $2.12 \times 10^3$ where increments became more pronounced. These results suggest that VBTR, with its superior shear capability, is better suited for forming smaller droplets under high Rei conditions, leading to larger mixing flux compared to NTR, thus having a higher operational limit range than NTR under identical processing capacity conditions.

The DSD of dispersed phase in NTR and VBTR is illustrated in Fig. 5 (c) and Fig. 5(d) for different $Re_i$. As $Re_i$ increases within the range of $1.59 \times 10^3$ to $3.71 \times 10^3$, both Taylor reactors exhibit a gradual shift towards higher values in DSD. The primary peak of VBTR's DSD demonstrates an initial increase followed by a decrease, while the regularity of the main peak in NTR's DSD appears less stable. At $Re_i$ equal to $1.59 \times 10^3$ and $3.71 \times 10^3$ respectively for both Taylor reactors, a secondary peak emerges within the range of 0.8 to 4.0 μm, indicating a transition from single-peak to double-peak and suggesting reduced uniformity in DSD due to increased fluid velocity or enhanced droplet coalescence in annular gap at lower flow rates as proposed by Qiao and Deng [55]. When the feed rate exceeds the minimum residence time required for shear dispersion in Taylor reactors, the DSD expands with increasing flow rate. For both Taylor reactors, the narrowest particle-size distributions occur within $Re_i$ values range concentrated between $2.12 \times 10^3$ to $2.65 \times 10^3$, indicating that the particle sizes are most uniform and smallest under these conditions. However, the particle-size interval for VBTR is significantly smaller than that for NTR, and overall particle-size distributions are narrower, indicating superior performance in terms of dispersion and stability.

### 4.2.3. *Effect of kerosene content*

At a constant rotational speed of 1800 rpm, $d_{3,2}$ values were obtained from the outlets of two different Taylor reactors under different operating conditions, as shown in Fig. 6(a). The observed changes in $d_{3,2}$ values align with findings reported by Tamhane et al. [56,57] for three different inlet Reynolds numbers ($Re_i = 1.59 \times 10^3$, $2.65 \times 10^3$, $3.71 \times 10^3$). For kerosene concentrations ranging from 1 % to 10 %, droplet sizes in VBTR are significantly smaller than those in NTR and increase with higher kerosene concentration. This suggests that a higher dispersed phase content results in more collisions between sheared droplets and increased droplet coalescence. When $Re_i = 3.71 \times 10^3$, all $d_{3,2}$ values in VBTR remain below 50 μm while NTR reaches a maximum value of 66 μm. At $Re_i = 1.59 \times 10^3$, the range of $d_{3,2}$ varies from 35 to 40 μm whereas at $Re_i = 3.71 \times 10^3$ it ranges from 47 to 66 μm. Therefore, the impact of kerosene concentration on droplet size cannot be ignored under high $Re_i$ conditions. The addition of vertical baffle plates results in far more tiny droplet sizes than NTR. Nevertheless, it does not decrease the likelihood of droplet collision. This suggests that the main

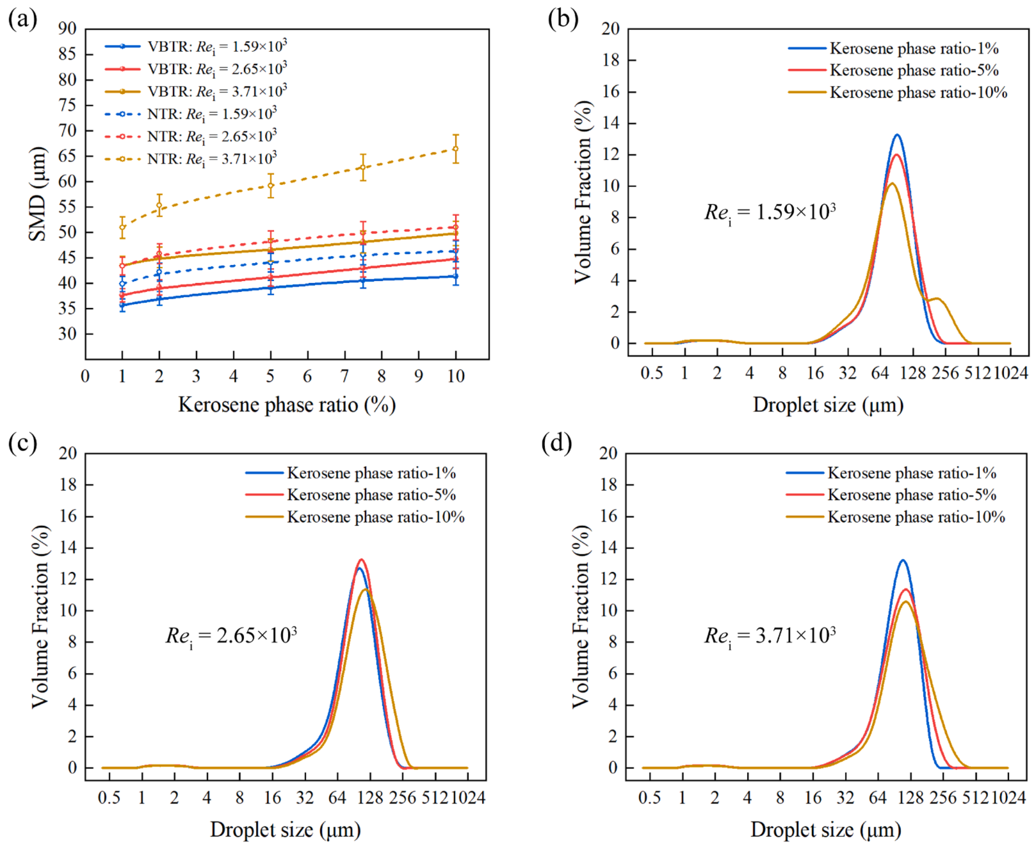


**Fig. 6.** (a) The effect of kerosene content on $d_{3,2}$ at three different $Re_i$; (b), (c) and (d) The DSD of different kerosene contents in the VBTR at three different $Re_i$ (data from experiments).

reason of the decrease in droplet size is the rise within turbulent kinetic energy and circumferential shear fragmentation frequency. At higher dispersed phase concentrations, both types of Taylor reactors exhibit varying degrees of droplet size increase. However, under high $Re_i$ conditions, VBTR shows a smaller increment, indicating more stable performance compared to NTR.

Fig. 6(b), Fig. 6(c) and Fig. 6(d) illustrates the DSD of VBTR at different kerosene concentrations. As the concentration of kerosene decreases, the $d_{50}$ value generally shifts with a decrease in droplet sizes, while the maximum droplet size content increases from 10.5 % to 14 %. This suggests a narrowing of the DSD and a reduction in the mean droplet size. However, under high dispersed phase concentration conditions ($Re_i = 1.59 \times 10^3$), the droplet size exhibits a bimodal distribution, with a negative correlation between the main peak size and kerosene concentration. The enlargement of droplets is primarily due to increased collision and coalescence probabilities within interfacial gaps as the dispersed phase content increases. Structural enhancements in mixing cannot mitigate the increased phase content leads to increased coalescence efficiency. For VBTR, under low turbulent conditions, secondary coalescence after droplet shearing also significantly affects droplet distribution and should not be disregarded.

### *4.3. Mixing performance evaluation*

The dispersed phase's volume fraction distribution characteristics within radial sections Z1-Z5, the outlet section, and the axial R-X plane ($z = 0$) were obtained via the field of flow depicted in Fig. 7(a). It is evident that NTR exhibits inadequate mixing performance. The continuous phase is unevenly distributed throughout the flow field, with a significant portion intermingling with the dispersed phase. Regions of "oil clusters" primarily located in the middle region of the annular gap and near the barrate wall are formed by the dispersed phase, which increases droplet collision and coalescence probabilities. The excessive relative velocity of the continuous phase also results in insufficient blending on a macroscopic scale during the dispersed phase, resulting in rapid passage through the annular gap and reduced mixing efficiency [19]. The introduction of vertical rib structures enhances tangential resistance along the main flow direction by dissipating fluid kinetic energy and prolonging residence time for dispersed phases. This effectively addresses non-uniform mass transfer issues in NTR, thereby improving the area of face-to-face between the phases and improving mixing efficiency. From Fig. 7(a), it can be observed that regardless of increasing *Re*, VBTR achieves complete two-phase mixing without affecting uniform distribution within the annular gap.

The *COV* curve in the NTR annular gap demonstrates a decreasing trend followed by an increasing trend as the position transitions from Z1 to Outlet, as illustrated in Fig. 7(b). Additionally, when *Re* is increased to $1.30 \times 10^5$, upward fluctuations are observed at positions Z1-Z3. The *COV* values at any cross-sectional position within the annulus and at the Outlet all exceed 0.1, indicating incomplete mixing of the two-phase fluid within the NTR annulus, which aligns with the conclusion drawn from Fig. 7(a). In contrast, within VBTR, there is a declining trend during the flow rising process of *COV* curve within the annulus (as depicted in Fig. 7(c)). As droplets migrate towards Taylor reactor, their dispersion becomes more uniform across radial sections of Taylor reactor. Under different *Re* conditions, *COV* curves sharply decrease after position Z3 and throughout this process; *COV* values are consistently below 0.05, signifying complete mixing of two-phase fluid achieved. When $Re = 1.17 \times 10^5$, outlet *COV* value drops to 0.018 and *COV* curve remains relatively flat under this condition. However, as *Re* continues to increase to $1.56 \times 10^5$, outlet *COV* value rises above 0.02 instead of dropping down again suggesting a slight decrease in homogeneous mixing. The ideal *Re* for achieving a high degree of mixing homogeneity in VBTR is found to be around $1.17 \times 10^5$ corresponding

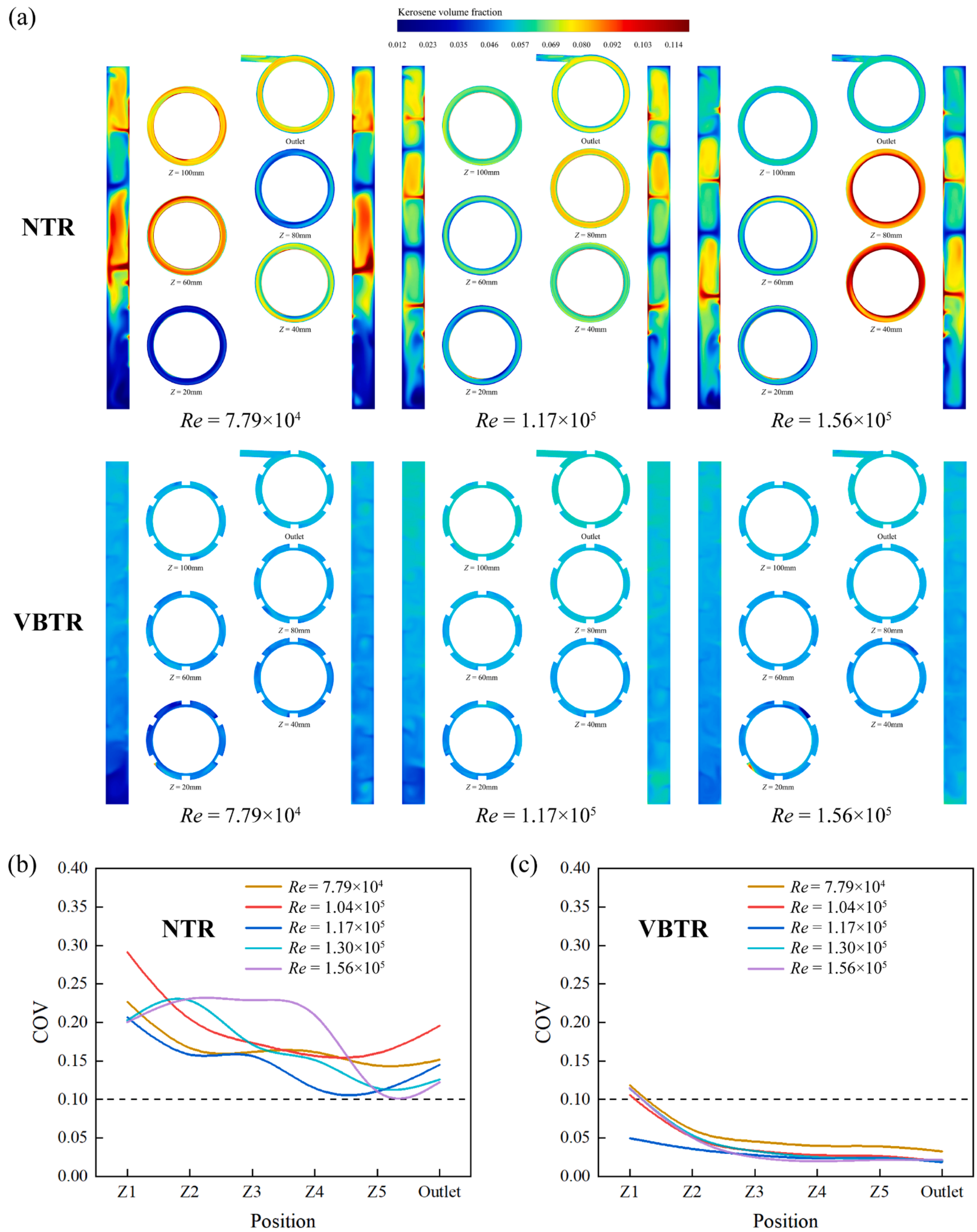


**Fig. 7.** (a) Dispersed phase volume fraction of two Taylor reactors; (b) The *COV* curves of NTR and (c) The *COV* curves of VBTR under different *Re* (data from numerical simulations).

rotational speed being set at 1800 rpm.

### 4.4. Droplet size distributions in the simulation

The dispersed phase $d_{3,2}$ value exhibits a decreasing trend in the outlet direction, as depicted in Fig. 8. The extent of small droplets continues to increase from the radial wall of the barrate, forming an interface at the corner of the vertical baffle. Droplet shear fragmentation primarily occurs near the barrate wall and at the edge of the vertical baffle plate. Through contrasting cloud photos captured in two different Reynolds number conditions (refer to Fig. 8(b) and Fig. 8(c)), It is evident that there is a decrease in DSD from 70 to 40 μm. At $Re = 1.17 \times 10^5$, droplet fragmentation mainly occurs in the middle part of annular gap (corresponding to height positions Z2-Z4), while at $Re = 1.56 \times 10^5$, it concentrates in the lower part of the annular gap (height positions Z1-Z3). Because of the tiny particles' quick dispersion within narrow spaces within the annular gaps means that neighboring positions near vertical ribs have a greater influence on tangential velocity at high rotating Reynolds numbers, leading to more pronounced inhibitory effects and a rapid decrease in $d_{3,2}$ value. These observations are consistent with the findings of Farzad and Kadam et al. [58,59]. The introduction of vertical baffles causes the concentration area for small droplets to shift towards liquid circumferential flow direction and leads to a more uniform distribution within annular gaps driven by Taylor vortex flow field.

### 4.5. Vorticity evaluation

When the Taylor reactor barrate rotates, Taylor-Couette flow develops in the annular gap fluid. As the rotational speed reaches a critical value, the centrifugal flow field in the annular gap becomes unstable. As

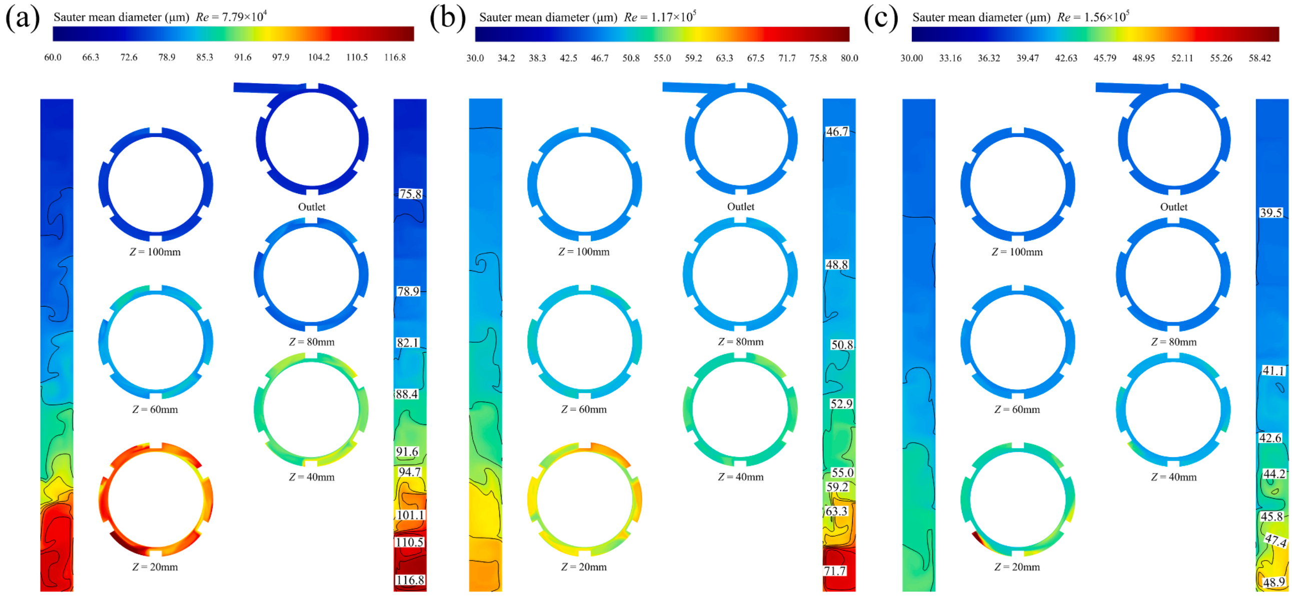


**Fig. 8.** Distributions of the $d_{3,2}$ at the VBTR: (a) $Re = 7.79 \times 10^4$; (b) $Re = 1.17 \times 10^5$ and (c) $Re = 1.56 \times 10^5$ (data from numerical simulations).

*Re* increases, the flow undergoes four distinct transitions: from laminar Couette flow to laminar Taylor vortices, then to wavy Taylor vortices, and finally to turbulent Taylor vortices. Fig. 9 illustrates the vortex cell patterns in two types of Taylor reactors under different *Re* conditions. As the rotational speed increases, the number of vortex cells gradually increases for both structures. In NTR, vortex cells are primarily

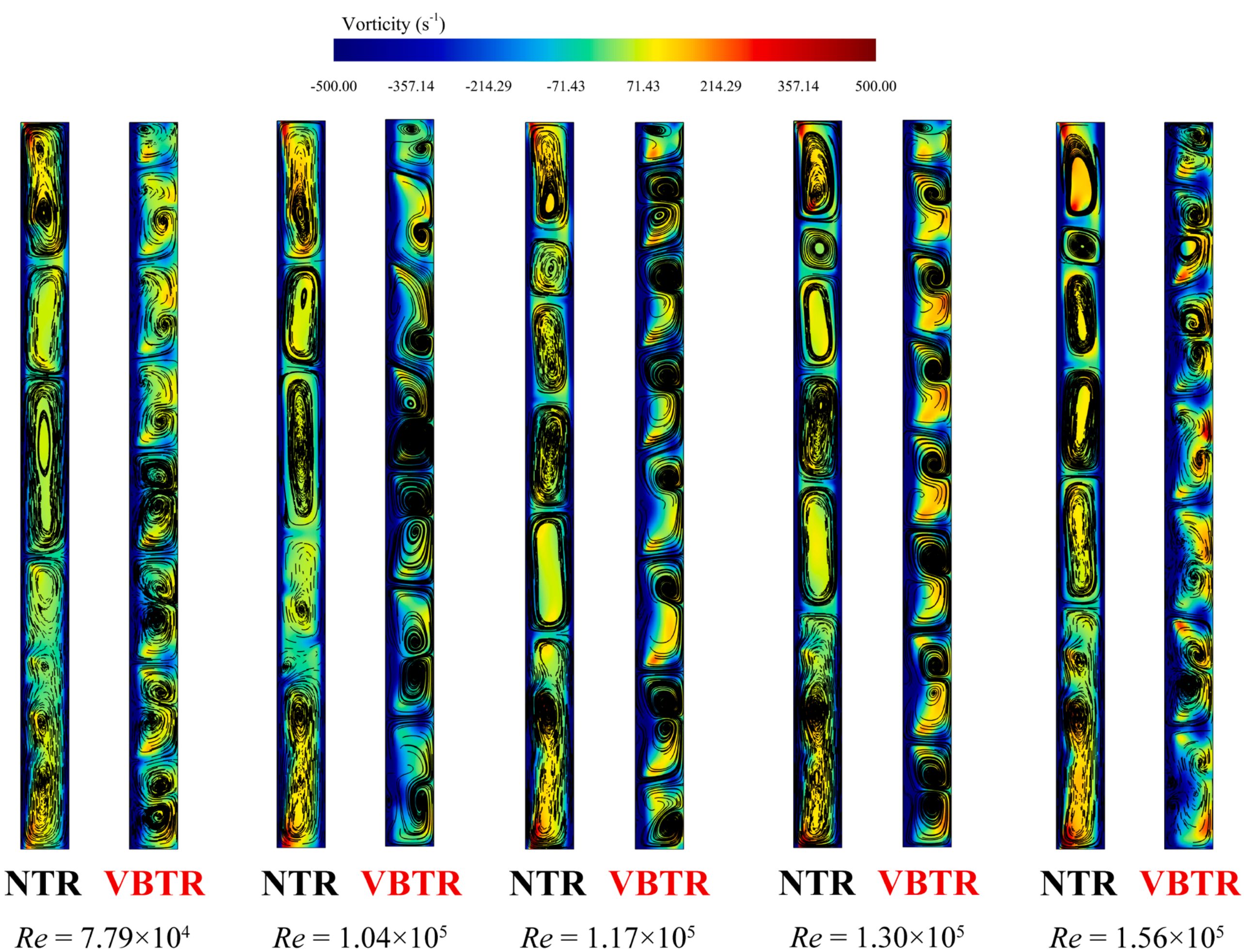


**Fig. 9.** Distributions of vortex cells and vortex streamlines at the VBTR and NTR under different *Re* (data from numerical simulations).

concentrated in the upper-middle part of the annular gap and their quantity remains constant; no vortex cells or effective Taylor vortices form in the lower part due to minimal influence from rotational speed on vortex generation. This indicates that increasing rotational speed does not significantly enhance mixing performance for this structure. In contrast, for VBTR, at low *Re* ($Re = 7.79 \times 10^4$), the entire annular gap is filled with vortex cells, fully transforming the flow into laminar Taylor vortices. All vortex cells exhibit a pair of counter-rotating vortices, with their quantity increasing and shapes becoming more regular as rotational speed rises. When *Re* exceeds $1.17 \times 10^5$, the number of vortex cells decreases and their strength weakens within R-X plane. The presence of vertical baffle structures does not promote the generation of vortices under high *Re* conditions. This can be attributed to the weakening effect of these vertical baffle plates on circumferential shear forces, resulting in a reduction in fluid circumferential motion intensity. Secondary flow phenomena typically require a certain level of circumferential flow intensity before they occur.

### 4.6. Velocity analysis

The entire system's speed vector variance and local cross-sections of two types of Taylor reactors at different *Re* are shown in Fig. 10. Comparing Fig. 10(b) and Fig. 10(d), VBTR's flow field shows a greater presence of velocity recirculation regions, due to unstable secondary flows caused by Taylor vortices, compared to the NTR flow field. At the same height on the axial R-X plane, point A in VBTR shows a $d_{50}$ value reduced by 34 μm compared to NTR, indicating that secondary flows in recirculation regions reduce droplet size. A similar evolution pattern of velocity vectors is observed within the annular gap of VBTR under different operating conditions. As fluid approaches the outlet, droplet size distribution becomes narrower and droplet size decreases. By comparing Fig. 10(a) and Fig. 10(b), it is evident that at $Re = 1.04 \times 10^5$, there are 8 vortex recirculation regions within the annular gap flow field; however, when *Re* increases to $1.30 \times 10^5$, this number significantly rises to 12, resulting in a difference of 20 μm in $d_{50}$ values for

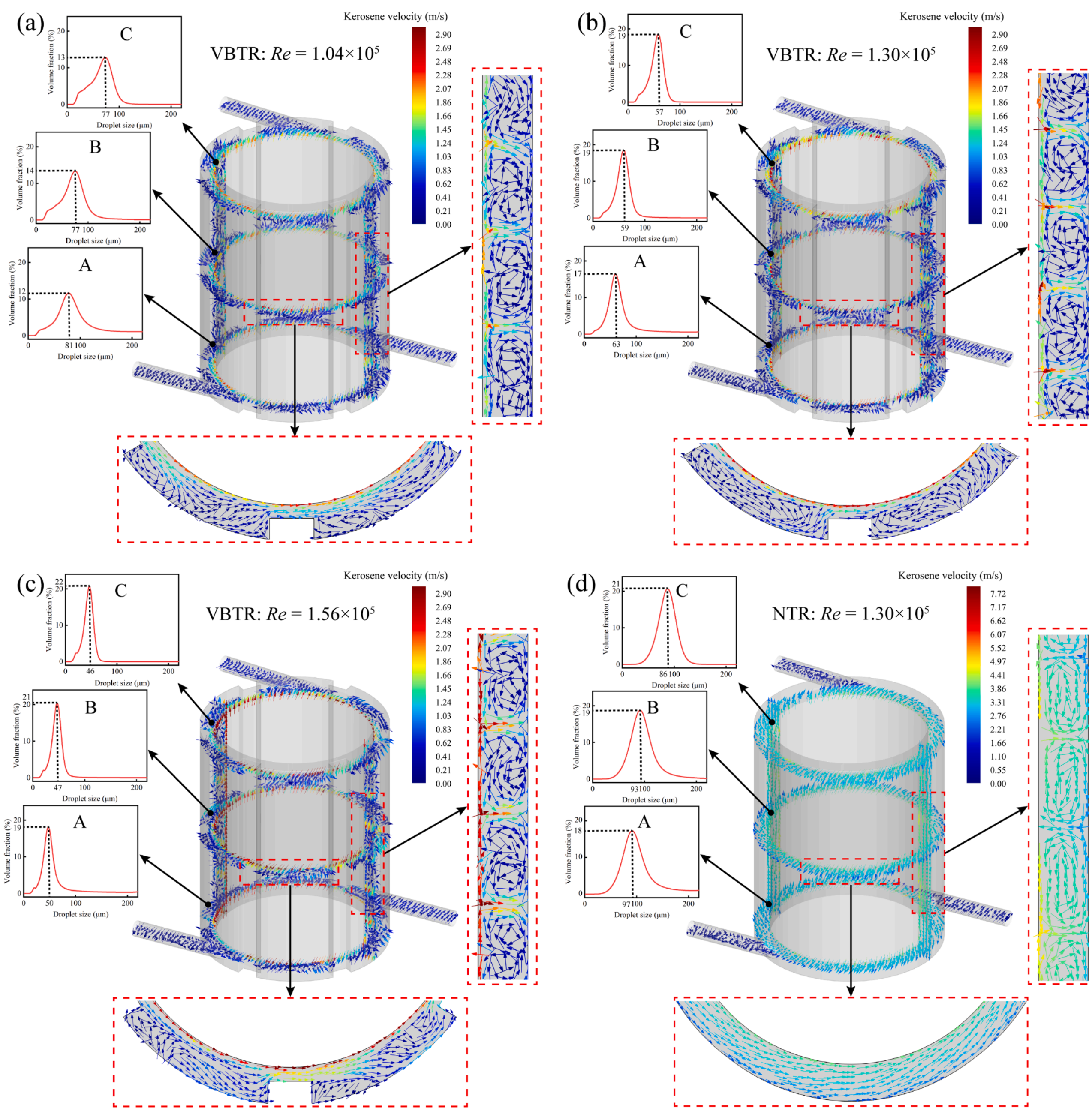


**Fig. 10.** Distributions of the $d_{50}$ and velocity vector at the VBTR and NTR under different *Re* (data from numerical simulations).

point C at equal heights between them. Similarly, comparing Fig. 10(b) and Fig. 10(c), no significant increase in the number of recirculation regions can be observed under both operating conditions; only one pair of recirculation regions is added when $Re = 1.56 \times 10^5$, but with an increase in circumferential velocity on circular sections which causes a decrease from 57 to 46 μm for droplet size at point C. This further confirms that unstable secondary flows play a crucial role in promoting droplet breakup.

### 4.7. Reynolds stress fluctuation

The Reynolds stress serves as a key indicator of turbulent stress in fluid flow and plays a significant role in the behavior of droplet fragmentation. The axial Reynolds stress consists of two Reynolds normal stresses, specifically $u'u'$ and $v'v'$, as well as a Reynolds shear stress, $u'v'$ [60]. Under the operating conditions are $Re = 1.17 \times 10^5$, $Re_a = 61.32$, and the kerosene concentration of 5 %, Fig. 11 illustrates the distribution of axial Reynolds stress in both Taylor reactors. Notably the influence of the annular gap fluid's Reynolds shear stress can be disregarded in both Taylor reactors, particularly NTR. A comparison of Fig. 11(a) and Fig. 11(d) reveals that VBTR exhibits an $u'u'$ magnitude exceeding 0.143 $m^2/s^2$ while NTR only reaches a maximum value of 0.12 $m^2/s^2$ for $u'u'$. This indicates that VBTR exhibits higher magnitudes and a broader stress distribution range. Flow velocity fluctuations occur near the barrate wall and close to baffle plates with periodic uniform changes along the entire height of annular gap (Z1-Z5) for VBTR (refer to Fig. 11(a) and Fig. 11(b)). In contrast, for NTR, there is a similar trend observed in distribution for both $u'u'$ and $v'v'$ (refer to Fig. 11(d) and Fig. 11(e)), primarily occurring near the outer side of annular gap; however, when $Z < 0.03$ or $Z > 0$ 0.1, the fluctuation in flow velocity can be neglected with earlier attenuation compared to VBTR. This phenomenon indicates that turbulent stresses are smaller in NTR than those in VBTR. Consequently, the vertical baffle plate enhances mixing within annular gaps which has been demonstrated to have several benefits, such as a strong rise in Reynolds stresses inside the flow field, a longer decay time, and a greater frequency of tiny droplet fragmentation.

### 4.8. Turbulent kinetic energy distribution

The turbulent kinetic energy distribution at different $Re$ is shown in

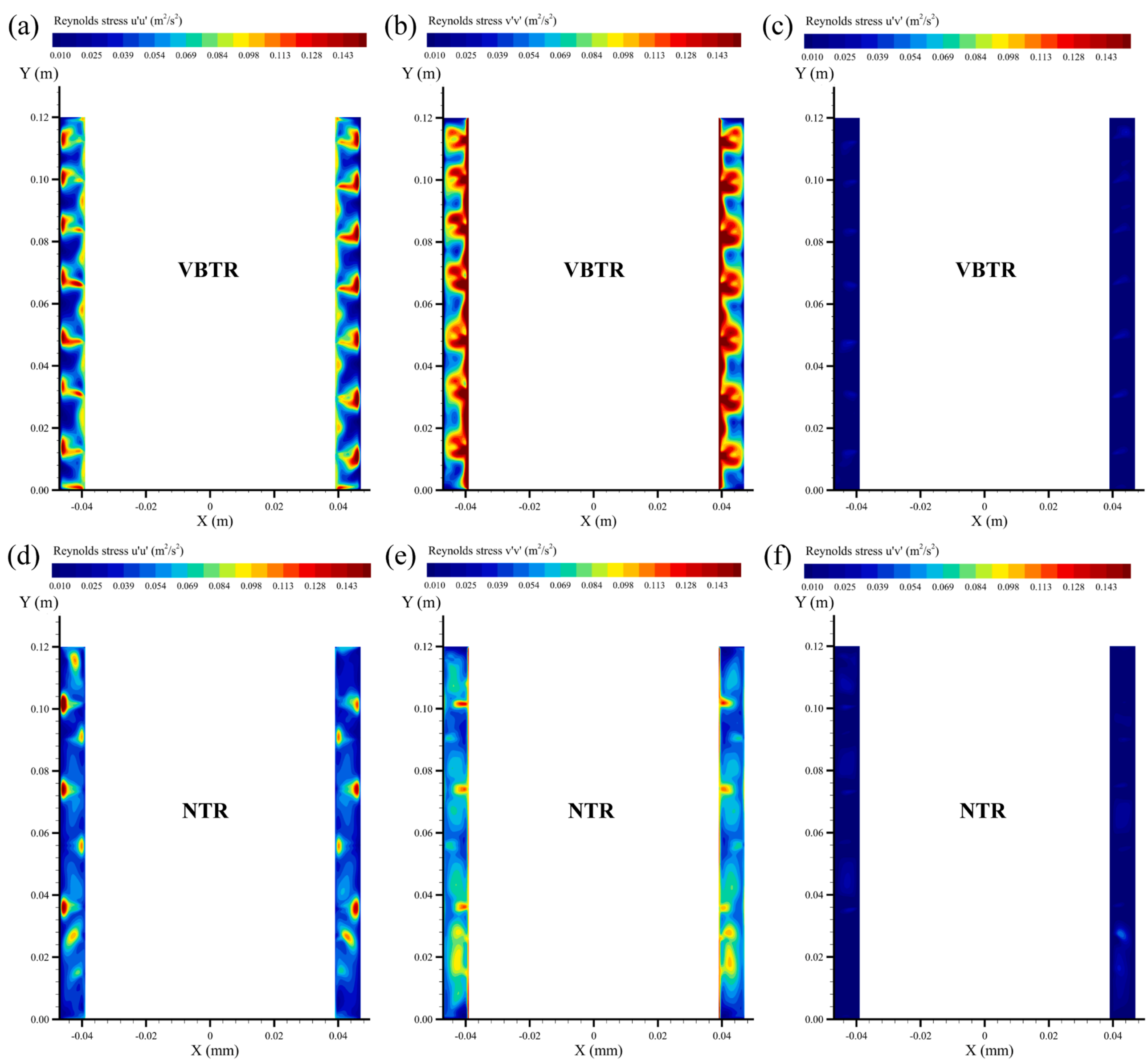


**Fig. 11.** The Reynolds stresses for axial direction in two Taylor reactors at $Re = 1.17 \times 10^5$ (data from numerical simulations).

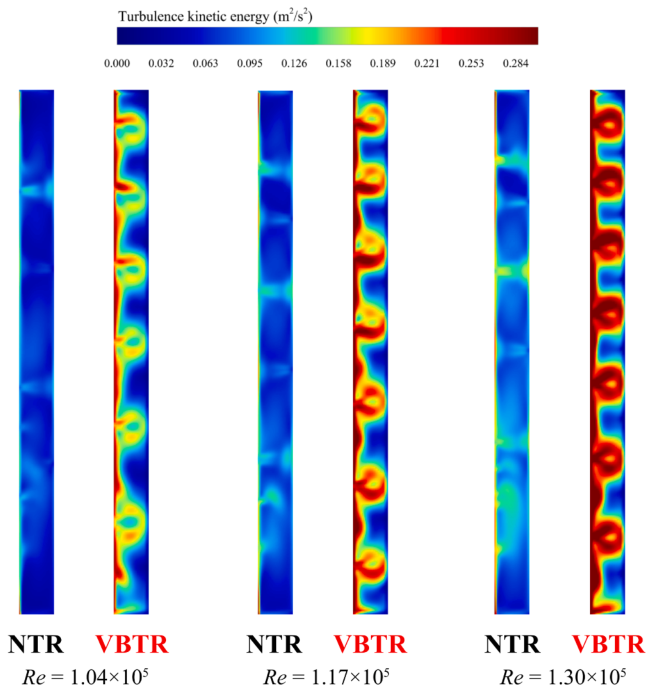


**Fig. 12.** The turbulence kinetic energy of two Taylor reactors under different *Re* (data from numerical simulations).

Fig. 12. When $Re \geq 1.17 \times 10^5$, both the spatial extent and maximum value of turbulent kinetic energy are elevated in VBTR. The turbulent kinetic energy distribution for VBTR predominantly concentrates within the barrate wall region and extends throughout the entire R-X plane along the axial direction, exhibiting periodic fluctuations that align with the findings presented in Section 4.7 on Reynolds stress fluctuation. Conversely, in NTR, turbulent kinetic energy primarily accumulates within the annulus and lower region. With increasing *Re*, the spatial extent of turbulent kinetic energy in VBTR expands rapidly. Conversely, while the distribution area of turbulent kinetic energy for NTR remains relatively stable, its turbulence intensity also undergoes some degree of augmentation. Consequently, designs such as VBTR result in an enlargement in the region characterized by high levels of turbulent kinetic energy within the annulus, thereby enhancing fragmentation of smaller droplets.

## 5. Conclusion

The inadequate blending and mass transfer performance observed in traditional Taylor reactors, particularly in systems with a high liquid-liquid phase ratio, is mainly attributed to their smooth internal and external wall surfaces. To address this issue and improve mixing efficiency, this study combines numerical simulations (CFD-PBM) with experimental analysis to systematically analyze and compare the effectiveness of incorporating vertical baffle plates within the annular gap. Additionally, the internal flow characteristics of two types of Taylor reactors are comprehensively examined. The study explores the patterns of droplet size distribution and coefficient of variation (*COV*) values under varying rotating Reynolds number (*Re*), shedding light on the underlying mechanisms responsible for improved mixing efficiency based on properties of the flow field. These research results provide a foundation for optimizing hydraulic structure design and promoting practical applications of Taylor reactors. The main conclusions of this study are as follows:

(1) The issue of low mixing efficiency, attributed to excessive velocity of the continuous phase and non-uniform droplet size has been successfully resolved. This has led to a significant improvement in the mixing performance within the annular gap of the Taylor reactor. When $Re = 1.17 \times 10^5$, VBTR demonstrates the narrowest range in particle size distribution and achieves optimal mixing efficiency, with a value for the coefficient of variation (*COV*) as low as 0.018 and SMD of 38.3 μm.
(2) When $Re_i$ at $1.59 \times 10^3$ and $3.71 \times 10^3$ respectively, the droplet size distribution shifts from unimodal to bimodal, suggesting a decrease in the uniformity of droplet size distribution. However, the droplet size range of VBTR is significantly narrower than in NTR, indicating superior dispersion and stability performance for the equipment. Within a concentration range from 1 to 10 % for dispersed phase, there exists a positive correlation between droplet size and concentration for VBTR.
(3) The analysis of the flow characteristics within the annular gap shows that the introduction of vertical baffle plates enhances mixing performance. This enhancement is attributed to several advantages such as increased vorticity and vortex intensity within the annular gap, extended contact and mass transfer time for droplets, improved likelihood of unstable secondary flows, as well as enhanced turbulent stress and turbulent kinetic energy within the flow field.

## CRediT authorship contribution statement

**Xiao Dong:** Writing – original draft, Methodology, Investigation, Formal analysis. **Shijie Yan:** Software, Methodology. **Xiaoyong Yang:** Writing – review & editing, Methodology. **Ningpu Liu:** Writing – review & editing, Conceptualization. **Shilong Du:** Visualization, Methodology. **Bingjie Wang:** Methodology, Conceptualization. **Fuwei Lv:** Software, Methodology. **Zhishan Bai:** Writing – review & editing, Supervision, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the research reported in this paper.

## Data availability

The authors do not have permission to share data.

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## Appendix

| **Nomenclature** | |
|---|---|
| $B_b$ | Birth rate of daughter particles for the droplet breakup process |
| $B_c$ | Birth rate of daughter particles for the droplet coalescence process |
| $C_1$—$C_4$ | Dimensionless constants of PBM |
| $C_i$ | Dispersed phase's level in a specific location on a cross-section of a mixing section |
| $\overline{C}$ | Average concentration |
| *CFD* | Computational Fluid Dynamics |
| *COV* | Coefficient of variation |
| $d$ | Annular gap width |
| $d_i$ | Inlet nozzle diameter |
| $d_o$ | Outlet nozzle diameter |
| $d_{10}$ | Particle size corresponding to 10 % cumulative particle count |
| $d_{3,2}$ | Sauter mean diameter |
| $d_{50}$ | Particle size corresponding to 50 % cumulative particle count |
| $d_{90}$ | Particle size corresponding to 90 % cumulative particle count |
| $D_b$ | Death rate of mother particles for the droplet breakup process |
| $D_c$ | Death rate of mother particles for the droplet coalescence process |
| $D_i$ | Barrate nominal diameter |
| $D_o$ | Outer cylinder diameter |
| *DSD* | Droplet size distribution |
| *EMMS* | Energy-Minimization Multi-Scale |
| $f_i$ | Volumetric fraction occupied by each individual droplet size |
| $L$ | Working length |
| $n$ | Vertical baffle quantity |
| *NTR* | No baffle Taylor Reactor |
| *PBM* | Population Balance Model |
| *QUICK* | Quadratic Upwind Interpolation for Convection Kinematics |
| $Re$ | Rotating Reynolds number |
| $Re_a$ | Axial Reynolds number |
| $Re_i$ | Inlet Reynolds number |
| *RSM* | Reynolds Stress Model |
| $T$ | Vertical baffle thickness |
| $Ta$ | Taylor number |
| $u$ | Mean flow rate in the inlet |
| *UDFs* | User-Defined Functions |
| *VBTR* | Vertical baffle Taylor Reactor |
| $w$ | Mean axial velocity across the annular gap region |
| $W$ | Vertical baffle width |
| $y^+$ | Dimensionless ratio of friction velocity |
| $\Delta y$ | Distance between the first layer grid node and the wall |
| ***Greek letters*** | |
| $\rho$ | Fluid density |
| $\rho_c$ | Density of the continuous phase |
| $\mu$ | Fluid viscosity |
| $\mu_c$ | Viscosity of the continuous phase |
| $\nu$ | Fluid kinematic viscosity |
| $\sigma$ | Standard deviation |
| $\Omega$ | Rotating speed |
| $\eta$ | Diameter ratio |
| $\Gamma$ | Aspect ratio |
| $\varepsilon$ | Dissipation rate of turbulence |
| $\xi$ | Droplet size variable |
| $\alpha_d$ | Dispersed phase volume fraction |
| $u_\tau$ | Friction velocity |

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